

Estimating Binding Affinity By Free-Energy Methods

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1 Introduction

Structure-based virtual screening and molecular docking are now routine tools in early-stage drug discovery, allowing the rapid evaluation of large libraries of compounds against biomolecular targets.[3, 13] However, classical scoring functions, such as based on empirical, knowledge-based, or force field based models, often provide only semi-quantitative estimates of binding affinity. They can often fail to rank ligands reliably across diverse chemical series.[9, 16] To address these limitations, more rigorous approaches based on statistical mechanics have been developed to compute binding free energies using molecular simulations. These free energy methods can be employed to refine docking poses, rescore candidates, and support lead optimization programs

In this chapter, we introduce free energy calculations for ligand binding, with an emphasis on methods that can be used to improve or complement docking results. We discuss the underlying thermodynamic and statistical mechanical principles, then discuss major class of methods, including end-state methods (such as MM-PBSA and MM-GBSA), linear interaction energy (LIE), and alchemical free energy methods such as free energy perturbation (FEP), and thermodynamic integration (TI). Particular attention is paid to the distinction between relative and absolute binding free energy (RBFE and ABFE) calculations, and to how these techniques can be integrated into practical docking-based workflows.

2 Thermodynamic background

2.1 Free energy and binding equilibria

The crux of accurately estimating protein–ligand binding finding is the standard binding free energy, $\Delta G_{\text{bind}}^\circ$, which is directly related to the equilibrium association constant K_{a} and the experimental dissociation constant K_{d} . For the binding reaction



the standard free energy of binding is given by

$$\Delta G_{\text{bind}}^\circ = -RT \ln K_{\text{a}} = RT \ln K_{\text{d}},$$

where R is the gas constant and T is the absolute temperature. Here, K_{a} or K_{d} is defined at a standard concentration C° , usually 1 M, so that the free energy is unambiguously defined up to standard state corrections.

Macroscopic binding constants are, in turn, related to microscopic partition functions. In the canonical ensemble at temperature T , the classical configurational partition function of a system with potential energy $U(\mathbf{x})$ is

$$Z = \int d\mathbf{x} \exp[-\beta U(\mathbf{x})],$$

where $\beta = 1/(k_{\text{B}}T)$ and k_{B} is Boltzmann’s constant. The corresponding Helmholtz free energy is

$$A = -k_{\text{B}}T \ln Z + \text{constant},$$

where the constant collects kinetic and quantum contributions that may cancel when differences between similar states are considered.[21, 4] Formally, the standard binding free energy can be expressed as the difference between the free energies of the complex and protein and ligand separately in the same standard state,

$$\Delta G_{\text{bind}}^\circ = G^\circ(\text{PL}) - G^\circ(\text{P}) - G^\circ(\text{L}).$$

Free energy methods for binding affinity prediction approximate this difference by simulating one or more of these states, or related alchemical or restrained states, and exploiting relationships between free energy and ensemble averages.

2.2 Thermodynamic cycles

Direct estimation of $\Delta G_{\text{bind}}^{\circ}$ by simulating the association process itself is challenging. This can be attributed to the complicated binding process that involves large changes in translational, rotational, and conformational degrees of freedom, as well as solvent reorganization. Instead, most approaches employ the thermodynamic cycles that connect the physical binding process to more convenient alchemical or decoupling steps. A frequently used construction is the *double decoupling* cycle for absolute binding free energies.[8, 7] In this approach, one considers turning off or on the interactions of the ligand with its environment in two contexts: in aqueous solution and in the binding site of the receptor. Denoting by λ , a coupling parameter that interpolates between a fully interacting state ($\lambda = 1$) and a non-interacting or *decoupled* state ($\lambda = 0$), one defines a cycle in which

1. the ligand is decoupled from solvent in bulk solvent (usually water),
2. the ligand is restrained and inserted, or removed, from the binding site,
3. the ligand is recoupled in the complex,
4. and standard state corrections are applied.

Summing the free energy changes around the cycle yields $\Delta G_{\text{bind}}^{\circ}$ in terms of alchemical transformation free energies that are more amenable to simulation. Related cycles can be constructed for relative binding free energies, wherein one ligand is alchemically transformed into another in both bound and unbound states. This latter approach works very well for ligands that are structurally similar.

3 Classification of free energy methods

A large number of computational techniques have been proposed to estimate binding free energies, differing in the level of theory, sampling strategy, and approximations used. The following broad classification is widely accepted:

- Level of theory:
 - End-state methods
 - Alchemical/pathway-based methods

- Sampling Strategy:
 - Relative binding free energy (RBFE)
 - Absolute binding free energy (ABFE)
- Level of approximation:
 - implicit solvent
 - Explicit solvent
 - Entropy contribution (present or absent)

3.1 End-state methods

End-state methods estimate the free energy difference between two states, e.g. bound and unbound, using the conformations sampled from equilibrium simulations of those states alone. These methods do not explicitly sample intermediate λ -states along an alchemical or physical pathway. Examples include continuum solvent approaches such as molecular mechanics Poisson–Boltzmann surface area (MM-PBSA) and generalized Born surface area (MM-GBSA), and linear interaction energy (LIE) models.[15, 11, 10] These methods are computationally less expensive than rigorous alchemical simulations. Therefore, attractive for processing large numbers of docking poses, however, they rely on significant approximations in the treatment of solvation and entropy.

3.2 Alchemical free energy methods

Alchemical methods define a non-physical path in the parameter space that continuously transforms one molecular system into another, e.g. ligand A into ligand B, or an interacting ligand into a non-interacting *ghost*, controlled by a coupling parameter λ . [4, 5] Free energy differences along this path are obtained using exact relationships from statistical mechanics, such as free energy perturbation (FEP), or thermodynamic integration (TI). These methods can, in principle, yield highly accurate relative or absolute binding free energies, provided that adequate sampling is achieved and the underlying force fields are reliable.

3.3 Relative v/s absolute binding free energies

Relative binding free energy (RBFE) methods estimate the difference in binding free energies between two ligands, A and B, for the same target:

$$\Delta\Delta G_{\text{bind}}^{\circ} = \Delta G_{\text{bind}}^{\circ}(\text{B}) - \Delta G_{\text{bind}}^{\circ}(\text{A}).$$

This is often achieved by alchemically transforming ligand A into ligand B in both the bound complex and in solution, and subtracting the two transformation free energies. Absolute binding free energy (ABFE) methods, by contrast, aim to compute $\Delta G_{\text{bind}}^{\circ}$ for a single ligand from first principles, usually via decoupling or insertion schemes coupled to restraints and standard state corrections.[8, 7, 5] In practice, RBFEs are widely used for lead optimization within a congeneric series, while ABFEs are attractive for ranking more diverse chemotypes or for assessing absolute potency.

4 End-state free energy methods

4.1 MM-PBSA and MM-GBSA

MM-PBSA and MM-GBSA combine molecular mechanics (MM) energies with implicit-solvent electrostatics (Poisson–Boltzmann (PB) or generalized Born (GB) models) and a nonpolar solvation term to estimate binding free energies from conformations obtained from MD trajectories.[15, 18, 11] For a protein–ligand complex, the binding free energy is approximated as

$$\Delta G_{\text{bind}} \approx \langle G_{\text{complex}} \rangle - \langle G_{\text{protein}} \rangle - \langle G_{\text{ligand}} \rangle,$$

where the angle brackets denote an ensemble average over conformations extracted from explicit-solvent simulations (or, in simpler protocols, minimized structures).

Each free energy term is decomposed as

$$G \approx E_{\text{MM}} + G_{\text{solv}} - TS_{\text{conf}},$$

with

$$E_{\text{MM}} = E_{\text{bonded}} + E_{\text{vdW}} + E_{\text{elec}},$$

and

$$G_{\text{solv}} = G_{\text{polar}} + G_{\text{nonpolar}}.$$

The polar solvation free energy, G_{polar} , is obtained either by solving the Poisson–Boltzmann equation (MM-PBSA) or using a GB approximation (MM-GBSA), while the nonpolar term is often modeled empirically as

$$G_{\text{nonpolar}} = \gamma \text{SASA} + b,$$

where SASA is the solvent-accessible surface area, and γ and b are empirical parameters.

The configurational entropy S_{conf} is estimated by normal-mode analysis, quasi-harmonic analysis, or other approximate schemes. However, in many applications it is neglected or assumed to cancel when comparing similar ligands. The resulting estimate of ΔG_{bind} is therefore best interpreted as a relative or semi-quantitative measure, particularly reliable when ranking closely related compounds.[11]

4.1.1 Equations for MM-PBSA

In a more explicit notation, the MM-PBSA estimate of the binding free energy can be written as

$$\begin{aligned} \Delta G_{\text{bind}}^{\text{MM-PBSA}} \approx & (\langle E_{\text{MM}}^{\text{complex}} \rangle + \langle G_{\text{solv}}^{\text{complex}} \rangle - T \langle S_{\text{conf}}^{\text{complex}} \rangle) \\ & - (\langle E_{\text{MM}}^{\text{protein}} \rangle + \langle G_{\text{solv}}^{\text{protein}} \rangle - T \langle S_{\text{conf}}^{\text{protein}} \rangle) \\ & - (\langle E_{\text{MM}}^{\text{ligand}} \rangle + \langle G_{\text{solv}}^{\text{ligand}} \rangle - T \langle S_{\text{conf}}^{\text{ligand}} \rangle). \end{aligned}$$

In practice, the conformational entropy term is often omitted or estimated from a reduced set of snapshots due to its high computational cost.

4.1.2 Advantages and limitations

The main advantages of MM-PBSA/GBSA are:

- They can reuse trajectories from classical MD simulations with explicit solvent, requiring only post-processing.
- They provide an interpretable decomposition of binding free energy into enthalpic and solvation contributions, and can be further decomposed into residue wise contribution.
- They are computationally cheaper than fully alchemical methods. Thus can be applied to tens or hundreds of complexes in docking or virtual screening programs.[11]

Limitations of these methods include:

- Dependence on the quality of the underlying force field and implicit-solvent models.
- Sensitivity to input structures, simulation length, and parameters, such as dielectric constants, atomic radii (Born radii), and ion concentration).
- Approximate treatment or neglect of configurational entropy and protein reorganization.

Despite these limitations, MM-PBSA/GBSA remain widely used as end-state rescoring tools to filter or re-rank docking poses and to support SAR.

4.2 Linear interaction energy (LIE)

The linear interaction energy (LIE) method is another end-state approach that estimates binding free energies from averaged interaction energies in bound and unbound simulations.[10] In its simplest form, the LIE equation for the binding free energy of a ligand is

$$\Delta G_{\text{bind}} \approx \alpha \langle V_{\text{elec}}^{\text{L-S}} \rangle_{\text{bound}} + \beta \langle V_{\text{vdW}}^{\text{L-S}} \rangle_{\text{bound}} + \gamma,$$

where $V_{\text{elec}}^{\text{L-S}}$ and $V_{\text{vdW}}^{\text{L-S}}$ are the electrostatic and van der Waals interaction energies between the ligand and its surroundings (solvent and protein), and α , β , and γ are empirical parameters. In some variants, differences between bound and unbound averages are used:

$$\Delta G_{\text{bind}} \approx \alpha (\langle V_{\text{elec}}^{\text{L-S}} \rangle_{\text{bound}} - \langle V_{\text{elec}}^{\text{L-S}} \rangle_{\text{unbound}}) + \beta (\langle V_{\text{vdW}}^{\text{L-S}} \rangle_{\text{bound}} - \langle V_{\text{vdW}}^{\text{L-S}} \rangle_{\text{unbound}}) + \gamma.$$

The parameters are calibrated against a training set of ligands with known experimental binding free energies for a given target class, which makes LIE semi-empirical. Because it relies on relatively short MD simulations and simple energy averages, LIE can be computationally efficient while still capturing important trends in binding.[10]

5 Alchemical free energy methods

5.1 Free energy perturbation (FEP)

Free energy perturbation is based on a relationship introduced by Zwanzig[21], which expresses the Helmholtz free energy difference between two states A

and B in terms of an ensemble average over configurations of state A. Let $U_A(\mathbf{x})$ and $U_B(\mathbf{x})$ be the potential energy functions of states A and B, respectively, defined over the same configurational space. The free energy difference $\Delta G_{A \rightarrow B} = G_B - G_A$ is given exactly by the Zwanzig equation:

$$\Delta G_{A \rightarrow B} = -k_B T \ln \langle \exp [-\beta (U_B(\mathbf{x}) - U_A(\mathbf{x}))] \rangle_A.$$

In practice, this average is estimated from configurations sampled from simulations of state A, either by MD or Monte Carlo method. The exponential reweighting in this equation makes FEP very sensitive to the overlap between the configurational ensembles of states A and B. If A and B are too dissimilar, the estimate becomes noisy and biased. This warrants the use of multiple intermediate states and more sophisticated estimators.

5.1.1 Multistate FEP and windows

To improve convergence, one must introduce a sequence of intermediate states parameterized by $\lambda_0 = 0, \lambda_1, \dots, \lambda_N = 1$, with potential energies $U(\mathbf{x}; \lambda)$ that smoothly morphs A into B. Adjacent states are chosen such that their ensembles overlap significantly, and free energy differences between neighbors are computed via FEP,

$$\Delta G_{\lambda_i \rightarrow \lambda_{i+1}} = -k_B T \ln \langle \exp [-\beta (U(\mathbf{x}; \lambda_{i+1}) - U(\mathbf{x}; \lambda_i))] \rangle_{\lambda_i},$$

then summed to obtain the total

$$\Delta G_{A \rightarrow B} = \sum_{i=0}^{N-1} \Delta G_{\lambda_i \rightarrow \lambda_{i+1}}.$$

This multistate FEP strategy underlies many implementations of relative free energy calculations in drug discovery, where it is often combined with soft-core potentials, enhanced sampling, and cycle closure methods to ensure convergence.[1, 20]

5.2 Thermodynamic integration (TI)

Thermodynamic integration was originally developed by Kirkwood[14]. It expresses the free energy difference between A and B as an integral over λ of the ensemble average of the derivative of the potential energy with respect

to λ . Defining the λ -dependent potential ($U(\mathbf{x}; \lambda)$) that interpolates between A and B, one obtains

$$\Delta G_{A \rightarrow B} = \int_0^1 \left\langle \frac{\partial U(\mathbf{x}; \lambda)}{\partial \lambda} \right\rangle_{\lambda} d\lambda.$$

In practice, this integral is evaluated numerically by performing simulations at a discrete set of λ -values and computing the ensemble average $\langle \partial U / \partial \lambda \rangle_{\lambda}$ at each point. Standard quadrature schemes such as trapezoidal rule, Simpson's rule, or Gauss-Legendre, are used to approximate the integral.[6] TI avoids the exponential average inherent in FEP, and can be more stable when the integrand varies smoothly with λ . However, the accuracy of TI depends mainly on the choice and density of λ -points and on adequate sampling at each state. Sharp changes in the integrand, for example near endpoints or phase transitions, require careful treatment, e.g. non-linear λ intervals or soft-core potentials.

5.3 Bennett acceptance ratio (BAR) and MBAR

The Bennett acceptance ratio method provides an optimal estimator for the free energy difference between two states using samples from both.[2] If one has ensembles of configurations from states A and B, with potential energies $U_A(\mathbf{x})$ and $U_B(\mathbf{x})$, BAR determines $\Delta G_{A \rightarrow B}$ as the solution of

$$\sum_{i=1}^{N_A} \frac{1}{1 + \exp [\beta (\Delta U_i - \Delta G_{A \rightarrow B} - C)]} = \sum_{j=1}^{N_B} \frac{1}{1 + \exp [\beta (\Delta U'_j + \Delta G_{A \rightarrow B} + C)]},$$

where $\Delta U_i = U_B(\mathbf{x}_i^A) - U_A(\mathbf{x}_i^A)$, $\Delta U'_j = U_A(\mathbf{x}_j^B) - U_B(\mathbf{x}_j^B)$, and C is an adjustable constant, often chosen near $\Delta G_{A \rightarrow B}$, to improve the numerical stability. This equation is typically solved iteratively. BAR can be viewed as a statistically optimal combination of forward and reverse FEP estimates and is less sensitive to poor overlap than single-direction FEP.[2, 6]

The multistate Bennett acceptance ratio (MBAR) extends BAR to multiple thermodynamic states, e.g. many intermediate λ -windows.[17] Given samples from several states with reduced potentials $u_k(\mathbf{x}) = \beta_k U_k(\mathbf{x})$, MBAR produces statistically optimal estimates of free energies f_k up to an additive constant, by solving a system of self-consistent equations:

$$\hat{f}_k = -\ln \sum_{n=1}^N \frac{\exp [-u_k(\mathbf{x}_n)]}{\sum_l N_l \exp [\hat{f}_l - u_l(\mathbf{x}_n)]},$$

where N_l is the number of samples from state l , and the sum over n runs over all configurations pooled from all states. MBAR has become a standard analysis tool for alchemical free energy simulations, enabling efficient reuse of data and rigorous uncertainty estimates.[17]

6 Relative and absolute binding free energy calculations

Relative binding free energy calculations consider a pair of ligands, A and B, binding to the same receptor. The quantity of interest is

$$\Delta\Delta G_{\text{bind}}^{\circ} = \Delta G_{\text{bind}}^{\circ}(\text{B}) - \Delta G_{\text{bind}}^{\circ}(\text{A}),$$

which directly relates to differences in potency.

A standard thermodynamic cycle for RBFE can be written as

$$\Delta\Delta G_{\text{bind}}^{\circ} = \Delta G_{A \rightarrow B}^{\text{bound}} - \Delta G_{A \rightarrow B}^{\text{solv}},$$

where $\Delta G_{A \rightarrow B}^{\text{bound}}$ is the alchemical free energy of transforming ligand A into ligand B in the binding site, and $\Delta G_{A \rightarrow B}^{\text{solv}}$ is the same transformation in bulk solvent.[20] Both transformation free energies are computed with FEP, TI, or BAR/MBAR (**Confirm this fact**) along a set of λ -windows.

In lead optimization programs, a network of pairwise transformations among a series of ligands can be constructed, and free energy methods are used to predict relative potencies[20, 1] and guide synthetic optimisation strategies. Consistency checks around cycles in the network, cycle closure, provide diagnostics of convergence and sampling quality.

Absolute binding free energy methods aim to compute $\Delta G_{\text{bind}}^{\circ}$ for a single ligand with respect to the standard state. The double decoupling approach is a widely used, where one constructs a cycle in which the ligand is decoupled or coupled from its environment in both the binding site and in bulk solvent, subject to restraints that define the bound state.[8, 7]

A typical double-decoupling cycle is:

$$\Delta G_{\text{bind}}^{\circ} = \Delta G_{\text{dec}}^{\text{solv}} - \Delta G_{\text{dec}}^{\text{bound}} + \Delta G_{\text{rest}} + \Delta G_{\text{std}},$$

where $\Delta G_{\text{dec}}^{\text{solv}}$ and $\Delta G_{\text{dec}}^{\text{bound}}$ are the free energies of decoupling the ligand in solvent and in the binding site, respectively, ΔG_{rest} accounts for the introduction and removal of positional/orientational restraints, and ΔG_{std} corrects for the change from the simulation box concentration to the standard

concentration. The decoupling free energies are obtained through alchemical transformations using FEP or TI, while restraint and standard state corrections can be calculated analytically from statistical mechanics, often using harmonic or flat-bottom potentials.[8, 7, 5] Although computationally more demanding than RBF, ABFE techniques provide a direct link to experimental binding constants. Therefore, in principle, it can be applied across diverse chemical scaffolds.

7 Practical considerations for alchemical calculations

7.1 Force fields and solvent models

The reliability of free energy calculations depends largely on the accuracy of the underlying molecular mechanics force fields and solvent models. Most biomolecular force fields such as AMBER, CHARMM, OPLS, are parameterized to reproduce a variety of thermodynamic and structural data. However, imperfections can still introduce systematic errors, especially for challenging chemotypes, metal centers, or unusual protonation states.[19] Explicit solvent, e.g. TIP3P, TIP4P-Ew, SPC/E water models, is commonly used for alchemical simulations, as it allows more realistic treatment of solvation and hydrogen bonding. Implicit solvent can be used for speed, however may lead to larger deviations from experiment.

7.2 Sampling and convergence

Sampling is one of the central challenges in free energy calculations. Binding processes often involve slow degrees of freedom, such as side-chain rearrangements, loop motions, or large-scale conformational changes, which may not be adequately sampled on the time scales accessible to routine MD. Insufficient sampling can lead to hysteresis between forward and reverse transformations, inconsistent cycle closure, and large statistical uncertainties.[1]

Strategies to improve sampling include:

- Careful selection of λ intervals, with more windows in regions where the system changes rapidly, e.g. near endpoints or when turning on strong electrostatics.

- Use of soft-core potentials to avoid singularities and large energy fluctuations when Lennard-Jones interactions are turned off or on.
- Enhanced sampling techniques, such as replica-exchange, Hamiltonian replica exchange, metadynamics, to accelerate exploration of conformational space to overcome large energy barriers.
- Longer simulations or multiple independent replicates per λ -window to reduce statistical noise.

7.3 Restraints and standard state corrections

When computing ABFEs, positional and orientational restraints are necessary to define the bound state and prevent the ligand from drifting away during decoupling. Common choices include harmonic restraints between ligand and protein reference atoms or flat-bottom distance and angle restraints.[8, 7] The free energy associated with these restraints must be accounted for analytically, and appropriate corrections must be made to convert from the effective simulation concentration to the standard 1 M reference state. The standard state correction associated with a restraining volume V_{rest} is typically of the form

$$\Delta G_{\text{std}} = -k_{\text{B}}T \ln \left(\frac{C^{\circ} V_{\text{rest}}}{1} \right),$$

where $C^{\circ} = 1/(1661 \text{ \AA}^3)$ in units of molecules per $\text{\AA}^3$. Proper treatment of these corrections is essential for comparing computed ABFEs with experimental binding free energies.

8 Integration of free energy methods with docking

8.1 Free energy-based rescoring of docking poses

A widely adopted strategy is to use docking primarily as a rapid generator of candidate poses and then apply more rigorous free energy methods to a reduced set of top-ranked complexes.[12, 16] Typical workflows include:

1. Dock a library of ligands to generate multiple poses per ligand.

2. Filter by docking score and visual inspection to select plausible poses.
3. Subject selected complexes to short explicit-solvent MD simulations to relax unrealistic contacts and allow local rearrangements.
4. Apply end-state methods (MM-PBSA/GBSA) or LIE to rescore the complexes and refine the ranking.

For a still smaller subset of promising ligands, alchemical RBFE or ABFE calculations can be performed to provide more quantitative predictions, often with errors on the order of 1–2 kcal/mol when best practices are followed.[20, 1] This hierarchical approach leverages the speed of docking and the rigor of free energy simulations in a complementary manner.

8.2 Lead optimization and SAR support

In medicinal chemistry programmes, free energy calculations are particularly powerful for:

- Ranking congeneric series of analogs that differ by relatively small chemical modifications.
- Interpreting structure–activity relationships and suggesting modifications that improve potency while maintaining desirable physicochemical properties.
- Decomposing contributions in to enthalpy and entropy components of binding, which can suggest optimization strategies, such as rigidification, desolvation, substituent placement.[8]

RBFEs are especially suitable for proposing relative potency shifts between analog pairs, while ABFEs can help benchmark entire series against experimental data or compare distinct chemotypes.

9 Summary

To be written

References

- [1] Matteo Aldeghi, Vytautas Gapsys, and Bert L de Groot. Accurate estimation of ligand binding affinity changes upon protein mutation. *ACS central science*, 4(12):1708–1718, 2018.
- [2] Charles H Bennett. Efficient estimation of free energy differences from monte carlo data. *Journal of Computational Physics*, 22(2):245–268, 1976.
- [3] Natasja Brooijmans and Irwin D Kuntz. Molecular recognition and docking algorithms. *Annual review of biophysics and biomolecular structure*, 32(1):335–373, 2003.
- [4] Christophe Chipot and Andrew Pohorille. *Free energy calculations*, volume 86. Springer, 2007.
- [5] John D Chodera and David L Mobley. Entropy-enthalpy compensation: role and ramifications in biomolecular ligand recognition and design. *Annual review of biophysics*, 42:121–142, 2013.
- [6] Anita de Ruiter, Stefan Boresch, and Chris Oostenbrink. Comparison of thermodynamic integration and bennett acceptance ratio for calculating relative protein-ligand binding free energies. *Journal of computational chemistry*, 34(12):1024–1034, 2013.
- [7] Yuqing Deng and Benoît Roux. Computations of standard binding free energies with molecular dynamics simulations. *The Journal of Physical Chemistry B*, 113(8):2234–2246, 2009.
- [8] Michael K Gilson and Huan-Xiang Zhou. Calculation of protein-ligand binding affinities. *Annu. Rev. Biophys. Biomol. Struct.*, 36(1):21–42, 2007.
- [9] Holger Gohlke, Manfred Hendlich, and Gerhard Klebe. Knowledge-based scoring function to predict protein-ligand interactions. *Journal of molecular biology*, 295(2):337–356, 2000.
- [10] Tomas Hansson, John Marelius, and Johan Åqvist. Ligand binding affinity prediction by linear interaction energy methods. *Journal of computer-aided molecular design*, 12(1):27–35, 1998.

- [11] Nadine Homeyer and Holger Gohlke. Free energy calculations by the molecular mechanics poisson- boltzmann surface area method. *Molecular informatics*, 31(2):114–122, 2012.
- [12] Tingjun Hou, Junmei Wang, Youyong Li, and Wei Wang. Assessing the performance of the molecular mechanics/poisson boltzmann surface area and molecular mechanics/generalized born surface area methods. ii. the accuracy of ranking poses generated from docking. *Journal of computational chemistry*, 32(5):866–877, 2011.
- [13] William L Jorgensen. The many roles of computation in drug discovery. *Science*, 303(5665):1813–1818, 2004.
- [14] John G Kirkwood. Statistical mechanics of fluid mixtures. *The Journal of chemical physics*, 3(5):300–313, 1935.
- [15] Peter Kollman. Free energy calculations: applications to chemical and biochemical phenomena. *Chemical reviews*, 93(7):2395–2417, 1993.
- [16] Julien Michel, Marcel L Verdonk, and Jonathan W Essex. Protein-ligand binding affinity predictions by implicit solvent simulations: a tool for lead optimization? *Journal of medicinal chemistry*, 49(25):7427–7439, 2006.
- [17] Michael R Shirts and John D Chodera. Statistically optimal analysis of samples from multiple equilibrium states. *The Journal of chemical physics*, 129(12), 2008.
- [18] Jayashree Srinivasan, Thomas E Cheatham, Piotr Cieplak, Peter A Kollman, and David A Case. Continuum solvent studies of the stability of dna, rna, and phosphoramidate- dna helices. *Journal of the American Chemical Society*, 120(37):9401–9409, 1998.
- [19] Junmei Wang, Romain M Wolf, James W Caldwell, Peter A Kollman, and David A Case. Development and testing of a general amber force field. *Journal of computational chemistry*, 25(9):1157–1174, 2004.
- [20] Lingle Wang, Yujie Wu, Yuqing Deng, Byungchan Kim, Levi Pierce, Goran Krilov, Dmitry Lupyan, Shaughnessy Robinson, Markus K Dahlgren, Jeremy Greenwood, et al. Accurate and reliable prediction of relative ligand binding potency in prospective drug discovery by way of

a modern free-energy calculation protocol and force field. *Journal of the American Chemical Society*, 137(7):2695–2703, 2015.

- [21] Robert W Zwanzig. High-temperature equation of state by a perturbation method. i. nonpolar gases. *The Journal of Chemical Physics*, 22(8):1420–1426, 1954.